(c) Explain the construction and working of SF_6 circuit breaker. 7

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(d) For a 132 kV system, the reactance and capacitance upto the location of the circuit breaker is 3Ω and $0{\cdot}015\ \mu F$ respectively. Calculate the following frequency of transient oscillation, the maximum value of restriking voltage across the contacts of the circuit breaker, the maximum value of RRRV. 7

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324612(24)

B. E. (Sixth Semester) Examination, Nov.-Dec. 2015

(Old Scheme)

(Elect. Engg. Branch)

POWER SYSTEM PROTECTION and SWITCHGEAR

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) from each question is compulsory. Attempt any two parts from parts (b), (c) and (d) of each question.*

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1. (a) Why reset value is less than pickup value? 2

(b) Derive the torque equation of induction relay. Draw and explain briefly the shaded pole and induction cup structure of this relay. 7

(c) Draw the schematic diagram of a reactance relay and Mho relay and explain in brief. How these relay function? 7

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(d) Explain with the help of neat diagram the working and construction of Buchholz relay. 7

2. (a) What are the various type of faults that occur in an alternator? 2

(b) Describe the differential bus bar protection. What are the limitation of this protective systems. How does it overcome the problems of saturated of CTs. 7

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(c) Why the rotor faults are detrimental to the alternator? How the rotor faults are protected? Draw the circuit diagram. 7

(d) What are possible causes of loss of excitation? Give any six causes only. How the alternator behaves during loss of excitation. 7

3. (a) What are the faults that are likely to occur in a transformer? 2

(b) Explain with the help of neat sketch the working

principle of carrier current protection scheme and its advantages. 7

(c) Explain a scheme of protection for a ring main. Give its limitation. 7

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(d) For a 10 MVA, 132 kV/6·6 kV pwer transformer with delta-star connection, obtain the number of turns each current transformer should have for the differential protection scheme to circulate a current of 5A in the pilot wires. 7

4. (a) What is static relay? 2

(b) Explain the concept of duality in static comparators, with phasor diagram. 7

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(c) Logically develop the design of reactance relay using sampling amplitude comparator. Give the block diagram of relay. 7

(d) What is the phase comparator? Describe coincidence type phase comparator. 7

5. (a) Compare fuse with circuit breaker on any two points. 2

(b) Explain the phenomenon of current chopping in circuit breaker. 7